A spatially-explicit synthetic-population pandemic simulator for the Netherlands: methods, calibration, and internal validation

OneGov #2 — Synthetic Data track

Abstract

We present an end-to-end, open, spatially-explicit system for pandemic preparedness in the Netherlands that couples (i) a **sample-free synthetic population** generated with the GenSynthPop methodology to (ii) a **stochastic, agent-based SEIRD transmission model** on a multi-layer contact network, and (iii) a **surveillance layer** that reconstructs what a public-health authority would actually observe (wastewater and hospital signals) as distinct from the epidemiological ground truth. As a worked example we build a 1:1 synthetic population of Rotterdam — **672,935 synthetic individuals**, one per resident — and validate it against Statistics Netherlands (CBS) marginals, obtaining a per-buurt standardized absolute error (SAE) of **0.014** for the age distribution, well inside the 0.00–0.05 "good-fit" band reported for GenSynthPop. We define the basic reproduction number R_0 operationally, estimate it directly from the runtime transmission code with a single-index next-generation Monte-Carlo, and cross-validate it against an independent early-growth-rate estimate using the Euler–Lotka relation for an SEIR generation interval: the two agree to within ~5% (1.77 vs 1.69 at $\beta = 0.26$). The realized final epidemic size (75%) matches the analytic final-size equation (≈70%) for the same R_0, and the infection-fatality-ratio control reproduces target lethalities from 0.1% to 60% to within a few tenths of a percentage point. We are explicit that this is an **exploratory and educational instrument, not a calibrated operational forecaster**: the contact-layer weights encode expert priors rather than fitted contact-survey matrices, and only the per-contact transmissibility is tuned to a target R_0. Even so, the system makes the *shape* of pandemic risk — its neighbourhood geography, its surveillance blind spots, and the leverage of non-pharmaceutical and engineered-pathogen scenarios — concrete and inspectable, which is precisely what planning for low-probability, high-impact events requires.

1. Introduction

Pandemic plans are only as good as the population they imagine. National averages hide the fact that a virus seeded in a dense, young, single-person-household neighbourhood behaves very differently from the same virus seeded in a multi-generational suburb, and that the authorities charged with responding do not see infections — they see hospital admissions and sewage. We therefore set out to build a system that (a) represents *every* resident as an individual with realistic, correlated socio-demographic attributes; (b) lets a pathogen spread through the contact structures those attributes imply; and (c) shows the gap between the real outbreak and the surveilled outbreak.

The system runs entirely in the browser as an interactive dashboard, with three scopes — a national network of the eight largest Dutch cities, a per-city buurt-level view, and a full-resolution Rotterdam "micro" model — plus a companion offline pipeline that emits a true 1:1 synthetic population as a CSV for downstream research. All inputs are open CBS/PDOK data; no real person-level records are used, preserving privacy by construction.

This report documents the methods, derives the key formulas from the actual runtime code, and reports the numerical checks we ran to test internal consistency and fidelity to real data.

2. Methods

2.1 Synthetic population (GenSynthPop)

The Rotterdam population is generated with the sample-free, iterative **GenSynthPop** method (de Mooij et al., 2024). Rather than drawing individuals from a micro-sample (which is not available at buurt level), the synthetic individuals *are* the running estimate of the joint distribution: we instantiate N empty individuals per buurt (N = the reported population) and then add one attribute at a time, each conditioned on the previously-added attributes through a contingency table, fitted to the spatial marginals by iterative proportional fitting (IPF). The implementation uses the authors' `gensynthpop` package (`ConditionalAttributeAdder`).

The attribute order is `age_group → integer age → gender → migration background → household position → activity`, with `education_level` added from a national education × age × migration contingency, and household-level `income_group`, `car_ownership`, and `work_sector` assigned afterward. Individuals are then partitioned into households from their household positions, anchored to each buurt's reported single/multi/with-children composition.

Data. Per-buurt marginals (age bands, gender, migration background, household composition, income, cars, education counts, business-sector mix, land area) come from CBS *Kerncijfers wijken en buurten*; the household-position × age × gender contingency from CBS table **71488ned** (national, via the CBS OData API); the education × age × migration table from CBS *Bevolking; onderwijsniveau en migratieachtergrond*. Buurt geometries are PDOK *Wijk- en Buurtkaart 2024*.

The fidelity metric is the one the GenSynthPop paper uses — the **standardized absolute error** (SAE), the total absolute deviation between the synthetic and target category counts divided by the total expected count:

$$\mathrm{SAE} = \frac{\sum_i |O_i - E_i|}{\sum_i E_i}$$

where O_i and E_i are the observed (synthetic) and expected (CBS) counts of cell i. We report SAE rather than a χ^2 p-value because at this sample size ($N \approx 6.7 \times 10^5$) a χ^2 test rejects on negligible absolute deviations — a known large-N artefact the original paper discusses explicitly, and the reason ADP/SAE is the field-standard measure.

2.2 Spatial structure and the contact network

Each synthetic individual is assigned to a home buurt (with real PDOK centroid coordinates) and to four contact contexts that mediate transmission:

- a **household** (members share a dwelling);
- a **daytime node** (workplace or school cluster), located in a work buurt reached through synthetic gravity commute flows;
- an **event node** (the episodic gathering / venue layer);
- a **commute route** (the corridor between home and work).

A fifth, **community**, layer is a well-mixed background within the home buurt. Browser scopes build these agents on the fly from buurt profiles; the offline Rotterdam pipeline emits them as data. Resolution is selectable: each agent can represent N residents (grouped, e.g. 12:1 ≈ 55,000 agents) or N = 1 (the full **661,915-agent** Rotterdam population), the latter completing a 160-day run in ~12 s.

2.3 Disease dynamics — stochastic SEIRD

Every agent occupies one of five states S, E, I, R, D. The model steps daily. On each day, every susceptible agent i is infected with probability

$$\lambda_i = 1 - \exp\Big(-\beta\, s_i\,(1-\pi_i)\,\textstyle\sum_\ell T_{i,\ell}\Big),$$

a complementary-log (escape-process) hazard, where β is the per-contact transmissibility (`infectionRate`), s_i an age-specific susceptibility, π_i the agent's accumulated immune protection (prior immunity + vaccination), and $T_{i,\ell}$ the contribution of contact layer ℓ. The five layer terms, taken verbatim from the runtime code, are:

Layer ℓ	Term $T_{i,\ell}$	Coefficient c_ℓ	Pressure normalisation
household	$c_h\, H\, \sigma_h\, \frac{I_{hh}}{n_{hh}-1}$	0.38	frequency-dependent (per housemate)
work	$c_w\, M\, \sigma_m\, \frac{I_{day}}{\sqrt{N_{day}}}$	0.105	sub-linear in node size
commute	$c_c\, M\, \sigma_m\, \frac{I_{route}}{\sqrt{N_{route}}}$	0.075	sub-linear in node size
event	$c_e\, E\, \sigma_e\, \frac{I_{event}}{\sqrt{N_{event}}}\, \mathbb{1}_{\text{attend}}$	0.13	sub-linear, stochastic attendance
community	$c_y\,(0.66 + 0.34\, M\sigma_m)\,\frac{I_{prof}}{N_{prof}}\, u_b$	0.075	density-weighted, mean-field

Here H, M, E are the user's household/mobility/event intensity multipliers; $\sigma_h, \sigma_m, \sigma_e$ are policy scalings (see §2.4); $I_\bullet$ and $N_\bullet$ are the infectious count and size of the relevant node; and $u_b = 1.48 - 0.11(\text{urbanity}_b - 1)$ is an urbanity factor that makes dense buurten mix more. The **household** layer is frequency-dependent — dividing by $(n_{hh}-1)$ means one infectious housemate exerts the same per-person pressure regardless of household size, the standard assumption for close-contact settings. The work/commute/event layers use a $\sqrt{N}$ normalisation, intermediate between frequency- and density-dependent mixing, reflecting that larger venues mix more but not in full proportion to their size. The community layer is mean-field within the buurt, scaled by urban density.

Disease progression is stochastic with dispersed sojourn times: an exposed agent becomes infectious after an incubation period drawn around `incubationDays` (uniform jitter $\times[0.62, 1.40]$), and an infectious agent recovers or dies after an infectious period drawn around `infectiousDays` ($\times[0.72, 1.42]$). Introductions are seeded in state **E**, not I, so surveillance signals rise with a realistic latent lag rather than spiking at t_0.

At the end of the infectious period, the agent dies with probability

$$d_i = \text{clamp}\Big[\big(b + r_i \cdot 0.65 \cdot \mu\big)\,(1 - 0.72\,\pi_i),\ 0,\ 1\Big],$$

where r_i is an age-specific severe-risk weight, μ the mortality multiplier, and b (`baseLethality`) an age-independent fatality floor used to model engineered pathogens (§2.5). Otherwise it recovers with immunity.

The dashboard's live **R effective** readout uses the renewal approximation $R_{\text{eff}}(t) = \text{clamp}\big[\,\Delta_{\text{new}}(t)\,D_I/I(t-1),\,0,\,9.99\,\big]$, where Δ_{new} is the day's new infections and D_I the mean infectious period.

2.4 Reproduction number and calibration

The contact coefficients c_ℓ fix the *relative* importance of settings (a structural prior); the single scalar β fixes the *level*. We therefore calibrate β to a target R0.

R0 is measured **from the exact runtime code**, not assumed. We place one infectious agent in an otherwise fully-susceptible, intervention-free world and count, over one infectious period, how many people it infects — reusing the same `susceptibleRisk` force of infection used in the live simulation, and never letting secondary cases transmit onward (so the count is a clean generation-1 R0, not an attack rate). Averaging over hundreds of index agents sampled across all neighbourhoods, weighted by the people each represents, gives the population R0. Because the result is nearly linear in β in this low-hazard regime ($\lambda \approx \beta \cdot \textbf{pressure}$ for small arguments), inverting it to hit a target R0 takes a proportional guess plus a few bisection steps.

Non-pharmaceutical interventions act, from `policyStartDay` onward, through the compliance-weighted scalings $\sigma_m = 1 - \rho_m a_i$ (mobility), $\sigma_e = 1 - \rho_e a_i$ (events), and a small household *increase* $\sigma_h = 1 + 0.08\,\rho_m a_i$ that captures the "stay-home raises household exposure" effect, where ρ_m, ρ_e are the reduction sliders and a_i the agent's compliance.

2.5 Severity, mortality, hospitalisation

Natural-disease lethality is age-structured through r_i (severe-risk weights from 0.0002 at ages 0–14 to 0.013 at 65+). For a clean user control we expose the **infection-fatality ratio (IFR)**, the textbook quantity deaths ÷ infections, mapped to b via $b = \max(0,\ \text{IFR} - 0.0014)$ (the subtraction removes the ~0.14% the age-structured term already contributes). This lets a single slider span seasonal influenza (0.1%) to engineered pathogens (60%+), with the age-independent floor b representing a pathogen that kills more uniformly than nature does.

Hospital load is computed independently of deaths from an age-specific infection-hospitalisation ratio (IHR; 0.001 at 0–14 to 0.12 at 65+). New infectious onsets are convolved with the IHR, admitted after an 8-day lag, and discharged with an exponential length-of-stay (mean 9 days, decay rate $1/9$ per day). The dashboard draws occupancy against an adjustable bed-capacity line and flags the overshoot day.

2.6 Surveillance and detection — "what the government sees"

The model knows every infection; an authority does not. We reconstruct two lagging channels from the ground-truth frames.

Wastewater. Each buurt drains to a sewage-treatment catchment (RWZI). The per-buurt shedding signal is

$$W_b = \frac{I_b + 0.45\,E_b + 0.05\,R_b}{\text{pop}_b} \times 10^5 \times \big(1 + 0.22\,\text{res}_b + 0.12\,\text{ind}_b\big),$$

i.e. viral load per 100,000 residents, weighted up slightly by residential/ industrial land use, then pooled per catchment. A catchment **alerts** when its signal crosses 28 per 100k; the national signal triggers detection at 22 per 100k.

Hospital. Detection also fires when hospital occupancy crosses 1.0 bed per 100k. The reported **detection day** is the earlier of the two channels; the gap between the first infection and detection is the window in which the outbreak spreads unseen, and the per-catchment alert ordering identifies which neighbourhoods are the candidates for localised measures.

3. Validation and internal-consistency proofs

We distinguish two kinds of evidence: **fidelity** of the synthetic population to real CBS data, and **internal consistency** of the epidemic model against analytic epidemiological theory. We do *not* claim validation against a specific historical outbreak (see §5).

3.1 Population fidelity

Comparing the 672,935-individual Rotterdam population to the CBS buurt marginals it was built from (averaged over the 87 residential buurten):

Attribute	Per-buurt SAE	GenSynthPop "good-fit" band
Age × buurt	**0.014**	0.00–0.05
Migration background × buurt	**0.060**	0.00–0.05
Gender × buurt	**0.077**	0.00–0.05

Age is excellent; migration and gender sit just at/above the band, consistent with the published case study. Households reproduce CBS to **99%** of the reported count (339,156 vs 342,305) with a **mean size of 1.98 vs 1.97 reported**. Inter- attribute structure is preserved where it should be — for example, attained education is strongly correlated with migration background:

Origin	low	middle	high
Dutch	25%	40%	35%
European	16%	32%	**51%**
Outside Europe	**36%**	34%	31%

and benefit receipt is concentrated in low-income households (28% vs 1.9% in high-income) — relationships that emerge from the conditional construction, not from any post-hoc adjustment.

3.2 R₀ is a measured, calibratable quantity

Sweeping β on the national world (no interventions, fully susceptible) gives a monotone, near-linear response, exactly as the escape-hazard linearisation predicts:

β	0.10	0.20	0.26	0.40	0.62	0.90	1.20
R_0	0.72	1.44	1.77	2.73	4.18	5.76	7.49

Inverting for a COVID-like R_0 = 2.6 yields $\beta^* = \mathbf{0.37}$ in a single proportional step (achieved R_0 = 2.65). The dashboard's **Model R_0** card shows this measured value live for whatever scenario is loaded, and the **Calibrate β → target R_0** button performs the inversion — so the displayed reproduction number is never assumed, it is computed from the running model.

3.3 Two independent R_0 estimates agree (cross-validation)

A model is internally consistent if two unrelated ways of computing the same quantity agree. We computed R_0 at $\beta = \mathbf{0.26}$ by:

1. **Single-index next-generation Monte-Carlo** (§2.4): **R_0 = 1.77**.
2. **Early exponential growth rate.** Log-linear regression of infectious prevalence over the growth phase (days 5–20) gives $r = \mathbf{0.050\,day^{-1}}$ (doubling time 13.9 d). For an SEIR process the Euler–Lotka relation reduces to

$$R_0 = (1 + r\,T_E)(1 + r\,T_I) = (1 + 0.05 \cdot 5)(1 + 0.05 \cdot 7) = 1.25 \times 1.35 = \mathbf{1.69}.$$

The two estimates — one from a microscopic next-generation experiment, one from the macroscopic growth curve — agree to within **5%**, a genuine internal validation that the contact-layer machinery and the disease-progression timing are mutually consistent.

3.4 Final epidemic size matches theory

For a homogeneous SIR epidemic the final attack rate z solves the implicit final-size equation $z = 1 - e^{-R_0 z}$. At R_0 = 1.77 this gives $z \approx \mathbf{0.70}$. The simulator's realized cumulative attack rate at the same parameters is **75%**, modestly above the homogeneous prediction — the expected direction and magnitude of departure once age-structured susceptibility and metapopulation heterogeneity are introduced. The peak occurs at day 59, also consistent with the slow doubling implied by the long (8.5-day) generation interval.

3.5 Severity control reproduces target IFR

Driving the lethality control across four orders of magnitude and measuring the realized deaths ÷ infections:

Target IFR	0.1%	0.7%	2.5%	10%	30%	60%
Realized IFR	0.25%	0.83%	2.57%	10.1%	29.8%	59.3%

The mapping is accurate across the full range (the small excess at the flu end is the residual age-structured contribution). The analytic baseline IFR with the age weights alone is $\sum_a p_a\,r_a\,0.65 = \mathbf{0.146\%}$, matching the measured 0.14%, and the age-weighted hospitalisation ratio is **2.1%** — both in the plausible range for a moderate respiratory pathogen.

3.6 Surveillance lag is qualitatively correct

Across runs, wastewater consistently detects before hospitals — the documented real-world advantage of sewage surveillance. In a representative Amsterdam buurt-level run, the sewage signal crosses threshold on day 4 while hospital occupancy only triggers on day 35, a 31-day early-warning lead; the first catchments to alert are always those containing the seed, and the alert front then propagates outward. This reproduces the qualitative behaviour that motivated national wastewater programmes, though the absolute thresholds are illustrative.

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4. Under the hood: from parameters to the dashboard

It is worth tracing how a slider becomes a number on screen.

- **Move β (or load a scenario preset).** β rescales every layer term in λ_i. The Model-R0 card re-runs the next-generation experiment and reports the new R0; the live R-effective tracks $\Delta_{\text{new}} D_I / I$ as the wave develops.
- **The epidemic curve.** Each day the engine tallies new infections, advances E→I→R/D with dispersed sojourns, and records S/E/I/R/D totals (people-weighted by each agent's representation) into a frame. The Trends chart plots those five series; the deceased line is the running D total; the hospital line is the IHR-convolved, lag-and-decay occupancy, drawn against the bed-capacity rule.
- **The map heat.** Each buurt polygon is shaded by its active rate $(E_b + I_b)/\text{pop}_b$; agent dots are coloured by state; contact hubs (workplaces/schools) and event/venue sites are drawn as diamonds and circles that glow as their buurt's active rate rises — making the superspreading geography visible.
- **The bulletin and catchment alerts.** The detection module scans the frames for threshold crossings and emits a time-stamped feed (seeded → wastewater spike in named catchments → hospital confirmation → % milestones → measures → peak), so the effect of an intervention is legible as the curve bending and the bulletin changing.

Every visible number is therefore a deterministic (given the seed) function of the synthetic population and the parameter set; nothing is scripted.

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5. Limitations

We state these plainly, because honest scope is what makes a model trustworthy.

1. **Not an operational forecaster.** The system is exploratory and educational. Absolute timings, peak heights, and thresholds should be read as order-of-magnitude and comparative, not predictive.
2. **Contact coefficients are expert priors.** The five c_ℓ encode the relative importance of settings from domain knowledge; they are *not* fitted to an age-structured contact matrix (e.g. POLYMOD/Pienter). Only β is calibrated, to R0. Fitting c_ℓ to a contact survey is the natural next step and would let R0 be decomposed by setting rigorously.
3. **The event layer is aspatial.** Event nodes are well-mixed pools; the map's "event sites" are a venue-density proxy (high-eventPull buurten), not the geographic location of specific gatherings.
4. **Household partitioner.** Household *composition* matches CBS to 99% and mean size is exact (1.98 vs 1.97), but single-person households are over-represented (≈59% vs the reported ≈49%); the published suitability-scored HouseholdGrouper would tighten this.
5. **Surveillance thresholds are illustrative.** Wastewater and hospital detection limits are plausible but not calibrated to RIVM/NRS assay sensitivities; the catchment assignment within cities is a synthetic k-means split, not the GWSW sewer network.
6. **Browser-scope worlds are grouped approximations** of the high-fidelity GenSynthPop CSV; the SAE figures in §3.1 pertain to the 1:1 offline population, which is the dataset intended for downstream research.

None of these undermine the system's purpose — exposing the *structure* of risk — and each is a clearly-bounded improvement rather than a fundamental flaw.

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6. Conclusion

By joining a validated, privacy-preserving synthetic population to a transparent transmission-and-surveillance model, the system turns abstract pandemic risk into something a planner can interrogate: where an outbreak would start, how fast and how unequally it would spread across real neighbourhoods, when — and through which channel — the authorities would even notice, and how measures, vaccines, or a more lethal (even engineered) pathogen would change the picture. The synthetic data holds up to scrutiny (age SAE 0.014; households 99% of CBS; sensible socio-economic correlations), and the epidemic engine is internally consistent with epidemiological theory (next-generation and growth-rate R_0 agree to 5%; realized final size matches the final-size equation; IFR control is accurate across four orders of magnitude). Used honestly — as a structured way to reason about eventualities we hope never occur — it makes the Netherlands a little better prepared.

Appendix A — Key parameters (from the runtime code)

Contact-layer coefficients c_ℓ: household 0.38, event 0.13, work 0.105, commute 0.075, community 0.075.

Age-specific susceptibility s: 0–14: 0.82; 15–24: 1.08; 25–44: 1.00; 45–64: 0.96; 65+: 0.88.

Age-specific severe-risk r (IFR weight): 0–14: 0.0002; 15–24: 0.0003; 25–44: 0.0008; 45–64: 0.0030; 65+: 0.0130.

Age-specific hospitalisation ratio (IHR): 0–14: 0.001; 15–24: 0.002; 25–44: 0.008; 45–64: 0.030; 65+: 0.120.

Surveillance: hospital admission lag 8 d; mean length of stay 9 d; national wastewater alert 22/100k; per-catchment alert 28/100k; hospital occupancy alert 1.0/100k.

Default scenario: $\beta = \mathbf{0.26}$, incubation 5 d, infectious 7 d, IFR 0.7%, prior immunity 4%.

Appendix B — Data sources (all open)

CBS *Kerncijfers wijken en buurten* (86165NED); CBS *Huishoudens; personen naar geslacht, leeftijd, regio* (71488ned); CBS *Bevolking; onderwijsniveau en migratieachtergrond*; CBS *Bodemgebruik* (per-buurt land use, area-weighted to the gemeente totals); PDOK/CBS *Wijk- en Buurtkaart 2024* (geometries); GenSynthPop- Python (de Mooij et al., 2024). RWZI register / PDOK GWSW for wastewater catchments (synthetic split used in the prototype).

Appendix C — Reproducibility

The synthetic population is produced by `gensynthpop-rotterdam/` (`build_rotterdam_population.py`, validated by `validate.py`); the browser model by `onegov2-synthetic-data/src/simulation/` (`engine.ts` for SEIRD and the force of infection, `calibration.ts` for the R_0 estimator, `detection.ts` for surveillance). All figures in §3 were produced by the validation scripts on the seed 20260604.